Code

# Introduction to Linear Mixed-Effects Models: nlme Vs lme4 with Rstudio

## 1 Linear Mixed-Effect Model: Package nlme

Simply Speaking, a linear mixed-effect model is a linear additive model of multi-levels by groups, where the random effect within group is additive to the fixed effect between group.

$$\mathbf{Y} = \boldsymbol{X_f\beta} + \boldsymbol{Z_rU} + \boldsymbol{\Psi}$$

where

- $\boldsymbol{X_f}$ is the explanatory parameter with fixed effect of $\boldsymbol{\beta}$.
- $\boldsymbol{Z_r}$ is the explanatoryrandom parameter with random effects of $\boldsymbol{U}$.
- $\boldsymbol{U} \sim \mathcal{MVN}(0, \Sigma_Z)$ is to generate the random normal effect for $\boldsymbol{Z_r}$.
- $\boldsymbol{\Psi} \sim \mathcal{MVN}(0, \Sigma_R)$ is the residual distribution of Model $\mathbf{Y}$
- $\boldsymbol{Cov(\Sigma_U, \Sigma_R) = 0}$

$$\overbrace{\mathbf{Y}}^{\mathrm{M\,x\,1}} = \overbrace{\underbrace{\mathbf{X_f}}_{\mathrm{M\,x\,N}}\ \underbrace{\boldsymbol{\beta}}_{\mathrm{N\,x\,1}}}^{\mathrm{M\,x\,1}} + \overbrace{\underbrace{\mathbf{Z_r}}_{\mathrm{M\,x\,K}}\ \underbrace{\boldsymbol{U}}_{\mathrm{K\,x\,1}}}^{\mathrm{M\,x\,1}} + \overbrace{\boldsymbol{\Psi}}^{\mathrm{M\,x\,1}}$$

## 2 Restricted Maximum Likelihood (REML)

The fundemental concept of Restricted Maximum Likelihood (REML) is to estimate the variance residuals components by transformation of OLS the model to remove the fixed effects :

$$\mathbf{TY} = \boldsymbol{TX_f\beta} + \boldsymbol{TZ_rU} + \boldsymbol{T\Psi}$$

By finding a matrix of $\mathbf{T}$ in such a way to restrict the fixed effects parameter: $\boldsymbol{TX_f} = 0$, therefore,

$$\mathbf{TY} = \boldsymbol{TZ_rU} + \boldsymbol{T\Psi}$$

, where $\boldsymbol{TX_f\beta} = 0$.

That is, when $\boldsymbol{Y} \sim \mathcal{N}(X\beta, \Sigma_R)$ for the restriction of $\boldsymbol{TX_f} = 0$, then

$$\mathbf{TY} \sim \mathcal{N}(0, T\Sigma_R T').$$

The ML of the original equation:

$$\boldsymbol{-2l\,(\beta, V|Y, X) = (Y - X\beta)' V^{-1}(Y - X\beta) + log(|V|) + nlog(2\pi)}$$

, where variance-covariance matrix $V = Z\Sigma_U Z' + \Sigma_R$

The REML equation is:

$$\boldsymbol{-2l\,(V_r|Y, X) = Y'(T'V_r^{-1}T)Y + log(|V_r|) + nlog(2\pi)}$$

, where variance-covariance matrix $V_r = TVT' = T(Z\Sigma_U Z' + \Sigma_R)T' = TZ\Sigma_U Z'T' + T\Sigma_R T'$

- The way how to find $\boldsymbol{T}$ and how to estimate $\boldsymbol{\beta}, \boldsymbol{\Sigma_U}, \boldsymbol{\Sigma_R}$ can refer to the book of Shayle R. Searle (1992) and the paper of Harville (1977), Laird and Ware (1982), Pinero and Bates (2000), JJ (2006), Crawley (2013) and Satoh (2018)
- The advantage of REML approach over ANOVA
    - can produce unbiased estimates of variance and covariance parameters;
    - can analyze unbalanced designs; and
    - has a powerful prediction algorithm that extends the ideas in regression prediction algorithm to cover random as well as fixed effects.

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```
ratpupcsv <- read.table("ratpup.csv", header = TRUE, sep = ",", na.strings = "NA",
    dec = ".", strip.white = TRUE)
```

Copy CSV Search:

| | pup.id | weight | sex | litter | litsize | treatment |
|---|---|---|---|---|---|---|
| 1 | 1 | 6.6 | Male | 1 | 12 | Control |
| 2 | 2 | 7.4 | Male | 1 | 12 | Control |
| 3 | 3 | 7.15 | Male | 1 | 12 | Control |
| 4 | 4 | 7.24 | Male | 1 | 12 | Control |
| 5 | 5 | 7.1 | Male | 1 | 12 | Control |
| 6 | 6 | 6.04 | Male | 1 | 12 | Control |
| 7 | 7 | 6.98 | Male | 1 | 12 | Control |
| 8 | 8 | 7.05 | Male | 1 | 12 | Control |
| 9 | 9 | 6.95 | Female | 1 | 12 | Control |
| 10 | 10 | 6.29 | Female | 1 | 12 | Control |

Showing 1 to 10 of 322 entries Previous 1 2 3 4 5 … 33 Next

# 3 Descriptive statistics: ratpupcsv

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```
str(ratpupcsv)
```

```
'data.frame':   322 obs. of  6 variables:
 $ pup.id   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight   : num  6.6 7.4 7.15 7.24 7.1 6.04 6.98 7.05 6.95 6.29 ...
 $ sex      : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 1 1 ...
 $ litter   : int  1 1 1 1 1 1 1 1 1 1 ...
 $ litsize  : int  12 12 12 12 12 12 12 12 12 12 ...
 $ treatment: Factor w/ 3 levels "Control","High",..: 1 1 1 1 1 1 1 1 1 1 ...
```

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```
summary(ratpupcsv)
```

```
     pup.id           weight          sex          litter         litsize
  Min.   :  1.00   Min.   :3.680   Female:151   Min.   : 1.00   Min.   : 2.00
  1st Qu.: 81.25   1st Qu.:5.650   Male  :171   1st Qu.: 7.00   1st Qu.:12.00
  Median :161.50   Median :6.055                Median :13.50   Median :14.00
  Mean   :161.50   Mean   :6.081                Mean   :13.38   Mean   :13.33
  3rd Qu.:241.75   3rd Qu.:6.397                3rd Qu.:19.75   3rd Qu.:16.00
  Max.   :322.00   Max.   :8.330                Max.   :27.00   Max.   :18.00
    treatment
  Control:131
  High   : 65
  Low    :126
```

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```
mysummary <- function(x) c(N = length(x), MEAN = mean(x, na.rm = TRUE), SD = sd(x,
    na.rm = TRUE), MIN = min(x, na.rm = TRUE), MAX = max(x, na.rm = TRUE))
pivot_table_01 = summarize(ratpupcsv$weight, by = llist(ratpupcsv$treatment, ratpupcsv$sex),
    mysummary)
colnames(pivot_table_01)[1:3] = c("treatment", "sex", "weight")
```

Copy CSV Search:

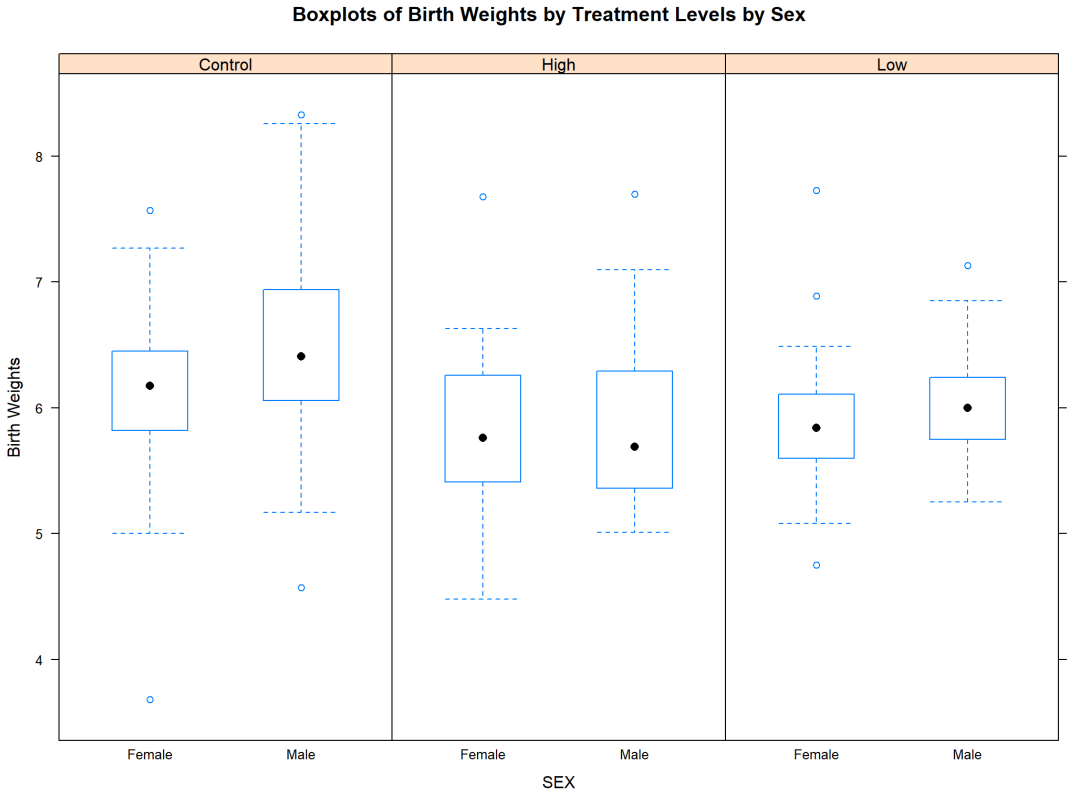
| | treatment | sex | weight | MEAN | SD | MIN | MAX |
|---|---|---|---|---|---|---|---|
| 1 | Control | Female | 54 | 6.1161 | 0.6851 | 3.68 | 7.57 |
| 2 | Control | Male | 77 | 6.4710 | 0.7538 | 4.57 | 8.33 |
| 3 | High | Female | 32 | 5.8516 | 0.6002 | 4.48 | 7.68 |
| 4 | High | Male | 33 | 5.9185 | 0.6909 | 5.01 | 7.7 |
| 5 | Low | Female | 65 | 5.8375 | 0.4505 | 4.75 | 7.73 |
| 6 | Low | Male | 61 | 6.0251 | 0.3803 | 5.25 | 7.13 |

Showing 1 to 6 of 6 entries Previous 1 Next

# Visualisations: ratpupcsv

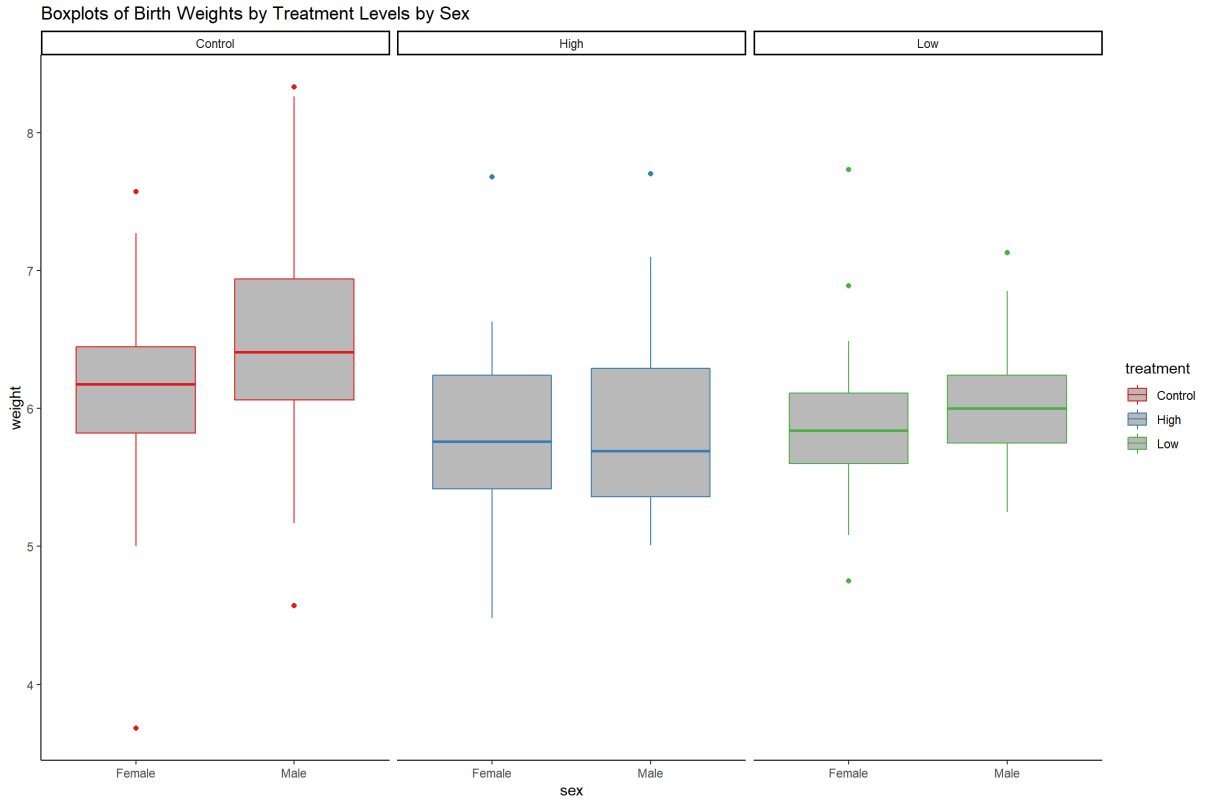
Hide

```
bwplot(weight ~ sex | treatment, data = ratpupcsv, aspect = 2, ylab = "Birth Weights",
    xlab = "SEX", main = "Boxplots of Birth Weights by Treatment Levels by Sex")
```




Hide

```
ggplot(ratpupcsv) + aes(x = sex, y = weight, colour = treatment) + geom_boxplot(fill = "#bdbdbd") +
    scale_color_brewer(palette = "Set1") + theme_classic() + facet_wrap(vars(treatment)) +
    ggtitle("Boxplots of Birth Weights by Treatment Levels by Sex")
```



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```
attach(ratpupcsv)
ranklit <- litsize + 0.01 * litter
sort(ranklit)
```

```
  [1]  2.12  2.12  3.23  3.23  3.23  4.03  4.03  4.03  4.03  8.25  8.25  8.25
 [13]  8.25  8.25  8.25  8.25  8.25  9.06  9.06  9.06  9.06  9.06  9.06  9.06
 [25]  9.06  9.06  9.26  9.26  9.26  9.26  9.26  9.26  9.26  9.26  9.26  9.27
 [37]  9.27  9.27  9.27  9.27  9.27  9.27  9.27  9.27 10.19 10.19 10.19 10.19
 [49] 10.19 10.19 10.19 10.19 10.19 10.19 10.22 10.22 10.22 10.22 10.22 10.22
 [61] 10.22 10.22 10.22 10.22 12.01 12.01 12.01 12.01 12.01 12.01 12.01 12.01
 [73] 12.01 12.01 12.01 12.01 12.13 12.13 12.13 12.13 12.13 12.13 12.13 12.13
 [85] 12.13 12.13 12.13 12.13 12.24 12.24 12.24 12.24 12.24 12.24 12.24 12.24
 [97] 12.24 12.24 12.24 12.24 13.05 13.05 13.05 13.05 13.05 13.05 13.05 13.05
[109] 13.05 13.05 13.05 13.05 13.05 13.10 13.10 13.10 13.10 13.10 13.10 13.10
 [ reached getOption("max.print") -- omitted 202 entries ]
```

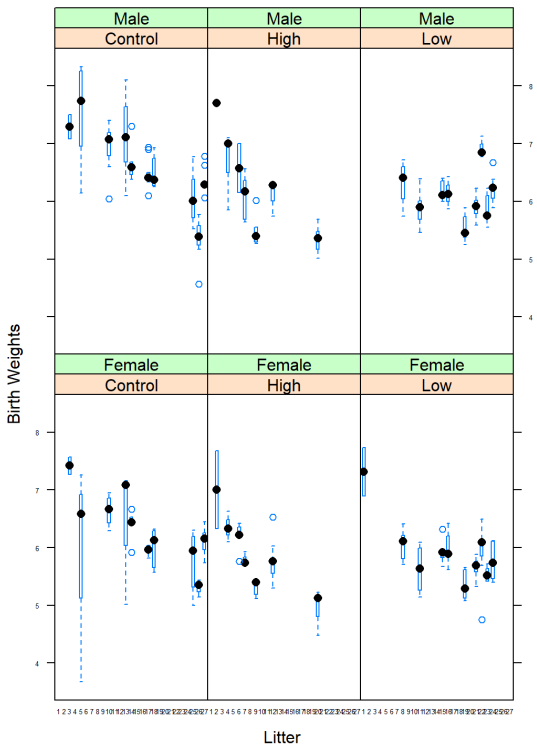
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```
ranklit <- factor(ranklit)
levels(ranklit) <- c("1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12",
    "13", "14", "15", "16", "17", "18", "19", "20", "21", "22", "23", "24", "25",
    "26", "27")
```

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```
bwplot(weight ~ ranklit | treatment * sex, data = ratpupcsv, aspect = 2, ylab = "Birth Weights",
    xlab = "Litter ", scales = list(cex = 0.4), main = "Boxplots of Birth Weights by Treatment levels by S
```

**Boxplots of Birth Weights by Treatment levels by Sex by Litter**



# 4 Modeling Specification Strategies:

## 4.1 Hypthesis Test 01 :

- to test the Random Effects (Litter)
- Approach: LR Test for Nested linear models
- $H_0 : \mu_{Litter} = 0$
- $H_A : \mu_{Litter} \neq 0$

### 4.1.1 lme: Model.01 with random effects(Litter)

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```
attach(ratpupcsv)
Model.01 <- lme(weight ~ treatment + sex + litsize + treatment * sex, random = ~1 |
    litter, method = "REML")
summary(Model.01)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC      BIC    logLik
  419.1043 452.8775 -200.5522

Random effects:
 Formula: ~1 | litter
        (Intercept) Residual
StdDev:   0.3106722 0.404337

Fixed effects: weight ~ treatment + sex + litsize + treatment * sex
                          Value  Std.Error  DF   t-value p-value
(Intercept)            7.911652 0.27496390 292 28.773422  0.0000
treatmentHigh         -0.799034 0.19429415  23 -4.112495  0.0004
treatmentLow          -0.383174 0.15967364  23 -2.399731  0.0249
sexMale                0.411688 0.07315410 292  5.627679  0.0000
litsize               -0.128382 0.01875336  23 -6.845819  0.0000
treatmentHigh:sexMale -0.107023 0.13176318 292 -0.812239  0.4173
treatmentLow:sexMale  -0.083866 0.10568189 292 -0.793568  0.4281
 Correlation:
                      (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmentHigh         -0.581
treatmentLow          -0.336  0.434
sexMale               -0.155  0.221  0.269
litsize               -0.910  0.372  0.045 -0.001
treatmentHigh:sexMale  0.105 -0.360 -0.150 -0.555 -0.020
treatmentLow:sexMale   0.141 -0.166 -0.345 -0.692 -0.036  0.385

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-7.47250744 -0.50014749  0.02911668  0.57348178  3.00962055

Number of Observations: 322
Number of Groups: 27
```

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```
anova.lme(Model.01, type = "sequential")
```

```
              numDF denDF  F-value p-value
(Intercept)       1   292 9093.772  <.0001
treatment         2    23    5.082  0.0149
sex               1   292   52.602  <.0001
litsize           1    23   47.374  <.0001
treatment:sex     2   292    0.466  0.6282
```

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```
anova.lme(Model.01, type = "marginal")
```

```
              numDF denDF  F-value p-value
(Intercept)       1   292 827.9098  <.0001
treatment         2    23   8.6905  0.0015
sex               1   292  31.6708  <.0001
litsize           1    23  46.8652  <.0001
treatment:sex     2   292   0.4656  0.6282
```

### 4.1.2 gls: Model.01A without random effects(Litter)

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```
library(Hmisc)
attach(ratpupcsv)
Model.01A <- gls(weight ~ treatment + sex + litsize + treatment * sex, data = ratpupcsv)
summary(Model.01A)
```

```
Generalized least squares fit by REML
  Model: weight ~ treatment + sex + litsize + treatment * sex
  Data: ratpupcsv
       AIC      BIC   logLik
  506.5099 536.5305 -245.255

Coefficients:
                         Value  Std.Error   t-value p-value
(Intercept)            7.900732 0.16229542  48.68118  0.0000
treatmentHigh         -0.807294 0.12051060  -6.69895  0.0000
treatmentLow          -0.381885 0.09273467  -4.11804  0.0000
sexMale                0.339576 0.08902602   3.81435  0.0002
litsize               -0.124188 0.01024669 -12.11978  0.0000
treatmentHigh:sexMale -0.178572 0.15325437  -1.16520  0.2448
treatmentLow:sexMale  -0.074357 0.12639429  -0.58829  0.5568

 Correlation:
                      (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmentHigh         -0.575
treatmentLow          -0.393  0.451
sexMale               -0.335  0.439  0.565
litsize               -0.907  0.372  0.092  0.014
treatmentHigh:sexMale  0.241 -0.700 -0.333 -0.582 -0.059
treatmentLow:sexMale   0.282 -0.328 -0.733 -0.705 -0.061  0.413

Standardized residuals:
        Min          Q1         Med          Q3         Max
-6.18740247 -0.50909449 -0.04113244  0.61273725  2.93938254

Residual standard error: 0.50151
Degrees of freedom: 322 total; 315 residual
```

### 4.1.3 Hypthesis Test 01 : results

- results indicated that reject the $H_0$ : $\mu_{Litter} = 0$
- therefore, $\mu_{Litter}$ is a significant random factor

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```
anova(Model.01A, Model.01)
```

```
          Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.01A     1  8 506.5099 536.5305 -245.2550
Model.01      2  9 419.1043 452.8775 -200.5522 1 vs 2 89.40562  <.0001
```

## 4.2 Hypthesis Test 02 :

- to test whether the residual variance differs between treatment groups

### 4.2.1 lme: Model.02

- with random effects(Litter)
- with a heterogeneous residual variance structure within treatment groups

Hide

```
attach(ratpupcsv)
Model.02 <- lme(weight ~ treatment + sex + litsize + treatment * sex, random = ~1 |
    litter, method = "REML", weights = varIdent(form = ~1 | treatment))
summary(Model.02)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC     BIC    logLik
  381.8847 423.163 -179.9423

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3134846 0.5147948

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | treatment
 Parameter estimates:
  Control       Low      High
1.0000000 0.5649830 0.6394383
Fixed effects: weight ~ treatment + sex + litsize + treatment * sex
                        Value  Std.Error  DF   t-value p-value
(Intercept)            7.937162 0.27736252 292 28.616565  0.0000
treatmentHigh         -0.808610 0.19628137  23 -4.119649  0.0004
treatmentLow          -0.390279 0.16301558  23 -2.394121  0.0252
sexMale                0.408131 0.09303486 292  4.386865  0.0000
litsize               -0.130007 0.01848708  23 -7.032332  0.0000
treatmentHigh:sexMale -0.094666 0.12919527 292 -0.732737  0.4643
treatmentLow:sexMale  -0.076013 0.10811858 292 -0.703053  0.4826
 Correlation:
                      (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmentHigh         -0.616
treatmentLow          -0.396  0.498
sexMale               -0.197  0.278  0.335
litsize               -0.896  0.378  0.070  0.000
treatmentHigh:sexMale  0.155 -0.361 -0.243 -0.720 -0.015
treatmentLow:sexMale   0.189 -0.248 -0.367 -0.860 -0.022  0.620

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.88670114 -0.52493419  0.02123518  0.57307286  2.56409983

Number of Observations: 322
Number of Groups: 27
```

Hide

```
anova(Model.01, Model.02)
```

```
         Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.01     1  9 419.1043 452.8775 -200.5522
Model.02     2 11 381.8847 423.1630 -179.9423 1 vs 2 41.21964  <.0001
```

### 4.2.2 Hypthesis Test 02 : results

- result indicated that the likelihood ratio testis significant (p<0.0001), so Model.02 with heterogeneous residual variances in the treatment groups has an optimal model fit, implying that the effects of heterogeneous residual variance structure exited within treatment groups.

## 4.3 Hypthesis Test 03 :

- to test whether the residual variances for the high and low treatment groups are equal

Hide

```
attach(ratpupcsv)
ratpupcsv$trtgrp[treatment == "Control"] <- 1
ratpupcsv$trtgrp[treatment == "Low" | treatment == "High"] <- 2
```

### 4.3.1 lme: Model.03

Hide

```
attach(ratpupcsv)
Model.03 <- lme(weight ~ treatment + sex + litsize + treatment * sex, random = ~1 |
    litter, method = "REML", weights = varIdent(form = ~1 | trtgrp))
summary(Model.03)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC      BIC    logLik
  381.0807 418.6065 -180.5404

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3145679 0.5147878

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5905488
Fixed effects: weight ~ treatment + sex + litsize + treatment * sex
                          Value  Std.Error  DF   t-value p-value
(Intercept)            7.942155 0.27838196 292 28.529705  0.0000
treatmentHigh         -0.809818 0.19550851  23 -4.142109  0.0004
treatmentLow          -0.390200 0.16383300  23 -2.381691  0.0259
sexMale                0.408195 0.09303539 292  4.387529  0.0000
litsize               -0.130383 0.01856367  23 -7.023574  0.0000
treatmentHigh:sexMale -0.092026 0.12461723 292 -0.738473  0.4608
treatmentLow:sexMale  -0.076397 0.10939797 292 -0.698337  0.4855
 Correlation:
                      (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmentHigh         -0.622
treatmentLow          -0.393  0.500
sexMale               -0.196  0.279  0.334
litsize               -0.897  0.382  0.068  0.000
treatmentHigh:sexMale  0.159 -0.351 -0.250 -0.747 -0.013
treatmentLow:sexMale   0.188 -0.247 -0.369 -0.850 -0.023  0.635

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.88725915 -0.52398492  0.01731335  0.56412339  2.55014733

Number of Observations: 322
Number of Groups: 27
```

Hide

```
anova(Model.03, Model.02)
```

```
         Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.03     1 10 381.0807 418.6065 -180.5404
Model.02     2 11 381.8847 423.1630 -179.9423 1 vs 2 1.196053  0.2741
```

### 4.3.2 Hypthesis Test 03 : results

- the non-significant result (p=0.2741) indicated that null hypothesis (the residual variances for the high and low dose treatments are equal) is statistically accepted, thereby accepting Model.03 better than Model.02

## 4.4 Hypthesis Test 04 :

- Is the residual variance for the combined high/low treatment group equal to the residual variance for the control group?
- with LR Test

Hide

```
anova(Model.01, Model.03)
```

```
         Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.01     1  9 419.1043 452.8775 -200.5522
Model.03     2 10 381.0807 418.6065 -180.5404 1 vs 2 40.02358  <.0001
```

### 4.4.1 Hypthesis Test 04 : results

- the very significant result (p<.0001) indicated that Model.03(with the pooled heterogeneous residual variances) is better than Model.01

## 4.5 Hypthesis Test 05 :

- Can any non-significant fixed effects be moved like the interaction treatment:sex?

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```
anova(Model.03, type = "marginal")
```

```
              numDF denDF  F-value p-value
(Intercept)       1   292 813.9441  <.0001
treatment         2    23   8.6436  0.0016
sex               1   292  19.2504  <.0001
litsize           1    23  49.3306  <.0001
treatment:sex     2   292   0.3167  0.7288
```

### 4.5.1 Hypthesis Test 05 : results

- The non-significant interaction (p=0.7288) result indicated that the fixed effects of the $treatment * sex$ should be removed from Model.03

Hide

```
attach(ratpupcsv)
Model.03M <- lme(weight ~ treatment + sex + litsize, random = ~1 | litter, method = "REML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.03M)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC      BIC    logLik
  372.2784 402.3497 -178.1392

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3146374 0.5144324

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5889108
Fixed effects: weight ~ treatment + sex + litsize
                  Value  Std.Error  DF   t-value p-value
(Intercept)    7.984202 0.27296903 294 29.249478  0.0000
treatmentHigh -0.862268 0.18293359  23 -4.713556  0.0001
treatmentLow  -0.433663 0.15226167  23 -2.848140  0.0091
sexMale        0.343431 0.04204323 294  8.168531  0.0000
litsize       -0.130681 0.01855194  23 -7.044036  0.0000
 Correlation:
              (Intr) trtmnH trtmnL sexMal
treatmentHigh -0.613
treatmentLow  -0.355  0.464
sexMale       -0.051  0.006  0.035
litsize       -0.910  0.403  0.064 -0.043

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.97351495 -0.53695365  0.01508652  0.54234475  2.58286992

Number of Observations: 322
Number of Groups: 27
```

## 4.6 Hypthesis Test 06 :

- Is there an effect of treatment on birth weight?
- Remember to use maximum likelihood (ML) for mixed models with different fixed effects structures if anova() is used for nested models comparsion

### 4.6.1 REML

Hide

```
attach(ratpupcsv)
Model.06_WTrm <- lme(weight ~ treatment + sex + litsize, random = ~1 | litter, method = "REML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.06_WTrm)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC      BIC    logLik
  372.2784 402.3497 -178.1392

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.3146374 0.5144324

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5889108
Fixed effects: weight ~ treatment + sex + litsize
                  Value  Std.Error  DF   t-value p-value
(Intercept)    7.984202 0.27296903 294 29.249478  0.0000
treatmentHigh -0.862268 0.18293359  23 -4.713556  0.0001
treatmentLow  -0.433663 0.15226167  23 -2.848140  0.0091
sexMale        0.343431 0.04204323 294  8.168531  0.0000
litsize       -0.130681 0.01855194  23 -7.044036  0.0000
 Correlation:
              (Intr) trtmnH trtmnL sexMal
treatmentHigh -0.613
treatmentLow  -0.355  0.464
sexMale       -0.051  0.006  0.035
litsize       -0.910  0.403  0.064 -0.043

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.97351495 -0.53695365  0.01508652  0.54234475  2.58286992

Number of Observations: 322
Number of Groups: 27
```

Hide

```
Model.06_WNOTrm <- lme(weight ~ sex + litsize, random = ~1 | litter, method = "REML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.06_WNOTrm)
```

```
Linear mixed-effects model fit by REML
 Data: NULL
       AIC      BIC    logLik
  381.7586 404.3497 -184.8793

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.4381779 0.5150813

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5881025
Fixed effects: weight ~ sex + litsize
                Value  Std.Error  DF   t-value p-value
(Intercept)  7.211043 0.28139850 294 25.625733   0e+00
sexMale      0.348955 0.04213828 294  8.281194   0e+00
litsize     -0.099343 0.02212882  25 -4.489285   1e-04
 Correlation:
        (Intr) sexMal
sexMale -0.043
litsize -0.947 -0.035

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.81957536 -0.51656194  0.02117273  0.55878848  2.53064896

Number of Observations: 322
Number of Groups: 27
```

Hide

```
anova(Model.06_WNOTrm, Model.06_WTrm)
```

```
               Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.06_WNOTrm     1  6 381.7586 404.3497 -184.8793
Model.06_WTrm       2  8 372.2784 402.3497 -178.1392 1 vs 2 13.48015  0.0012
```

#### 4.6.1.1 ML

Hide

```
attach(ratpupcsv)
Model.06_WTrm <- lme(weight ~ treatment + sex + litsize, random = ~1 | litter, method = "ML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.06_WTrm)
```

```
Linear mixed-effects model fit by maximum likelihood
 Data: NULL
       AIC      BIC    logLik
  353.7734 383.9698 -168.8867

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.2884174 0.5137273

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5881703
Fixed effects: weight ~ treatment + sex + litsize
                  Value  Std.Error  DF   t-value p-value
(Intercept)    7.986093 0.25810690 294 30.941030  0.0000
treatmentHigh -0.864775 0.17148477  23 -5.042866  0.0000
treatmentLow  -0.434407 0.14242656  23 -3.050046  0.0057
sexMale        0.341896 0.04223590 294  8.094909  0.0000
litsize       -0.130701 0.01750951  23 -7.464595  0.0000
 Correlation:
              (Intr) trtmnH trtmnL sexMal
treatmentHigh -0.616
treatmentLow  -0.355  0.467
sexMale       -0.054  0.006  0.037
litsize       -0.911  0.407  0.065 -0.046

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.98845212 -0.52758888  0.02174839  0.55158792  2.59709135

Number of Observations: 322
Number of Groups: 27
```

Hide

```
Model.06_WNOTrm <- lme(weight ~ sex + litsize, random = ~1 | litter, method = "ML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.06_WNOTrm)
```

```
Linear mixed-effects model fit by maximum likelihood
 Data: NULL
       AIC      BIC    logLik
  368.3706 391.0179 -178.1853

Random effects:
 Formula: ~1 | litter
        (Intercept)  Residual
StdDev:   0.4209366 0.5146485

Variance function:
 Structure: Different standard deviations per stratum
 Formula: ~1 | trtgrp
 Parameter estimates:
        1         2
1.0000000 0.5871372
Fixed effects: weight ~ sex + litsize
                Value  Std.Error  DF   t-value p-value
(Intercept)  7.208088 0.27310606 294 26.392998   0e+00
sexMale      0.348624 0.04223474 294  8.254428   0e+00
litsize     -0.099158 0.02146517  25 -4.619464   1e-04
 Correlation:
        (Intr) sexMal
sexMale -0.044
litsize -0.947 -0.036

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-5.81693672 -0.52012175  0.01677768  0.56092789  2.54095453

Number of Observations: 322
Number of Groups: 27
```

Hide

```
anova(Model.06_WNOTrm, Model.06_WTrm)
```

```
                Model df      AIC      BIC    logLik   Test  L.Ratio p-value
Model.06_WNOTrm     1  6 368.3706 391.0179 -178.1853
Model.06_WTrm       2  8 353.7734 383.9698 -168.8867 1 vs 2 18.59723   1e-04
```

### 4.6.2 Hypthesis Test 06 : results

- The results with “ML” and “REML” indicated that the significant treatment effects is very significant in Model.06_WTrm (p< 1e-04)
- It should be aware that since Model.06_WTrm and Model.06_WNOTrm are under nested models specfication with one fixed parameter difference(treatment), therefore, the result of testing significance for both two approach are same. However, if their models ar not in nested specification, ML approach is more valid than the REML for two unnested model specification.

# 5 The first best model should be : Model.FB

Hide

```
attach(ratpupcsv)
Model.FB <- lme(weight ~ treatment + sex + litsize, random = ~1 | litter, method = "REML",
    weights = varIdent(form = ~1 | trtgrp))
summary(Model.FB)
```

```
 Linear mixed-effects model fit by REML
  Data: NULL
        AIC      BIC    logLik
   372.2784 402.3497 -178.1392

 Random effects:
  Formula: ~1 | litter
         (Intercept)  Residual
 StdDev:   0.3146374 0.5144324

 Variance function:
  Structure: Different standard deviations per stratum
  Formula: ~1 | trtgrp
  Parameter estimates:
         1         2
 1.0000000 0.5889108
 Fixed effects: weight ~ treatment + sex + litsize
                   Value  Std.Error  DF   t-value p-value
 (Intercept)    7.984202 0.27296903 294 29.249478  0.0000
 treatmentHigh -0.862268 0.18293359  23 -4.713556  0.0001
 treatmentLow  -0.433663 0.15226167  23 -2.848140  0.0091
 sexMale        0.343431 0.04204323 294  8.168531  0.0000
 litsize       -0.130681 0.01855194  23 -7.044036  0.0000
  Correlation:
               (Intr) trtmnH trtmnL sexMal
 treatmentHigh -0.613
 treatmentLow  -0.355  0.464
 sexMale       -0.051  0.006  0.035
 litsize       -0.910  0.403  0.064 -0.043

 Standardized Within-Group Residuals:
         Min          Q1         Med          Q3         Max
 -5.97351495 -0.53695365  0.01508652  0.54234475  2.58286992

 Number of Observations: 322
 Number of Groups: 27
```

Hide

```
anova(Model.FB, type = "marginal")
```

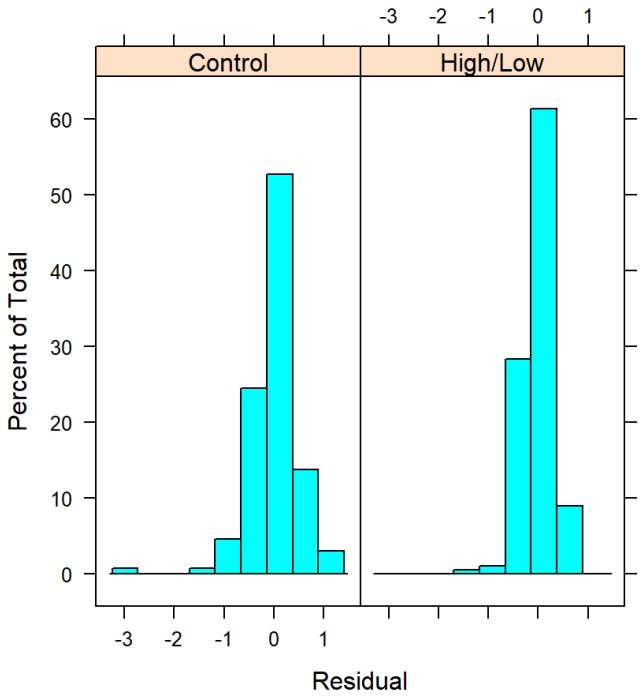
```
             numDF denDF  F-value p-value
 (Intercept)     1   294 855.5320  <.0001
 treatment       2    23  11.3870   4e-04
 sex             1   294  66.7249  <.0001
 litsize         1    23  49.6184  <.0001
```

# 6 Model diagnostics

## 6.1 Histogram

Hide

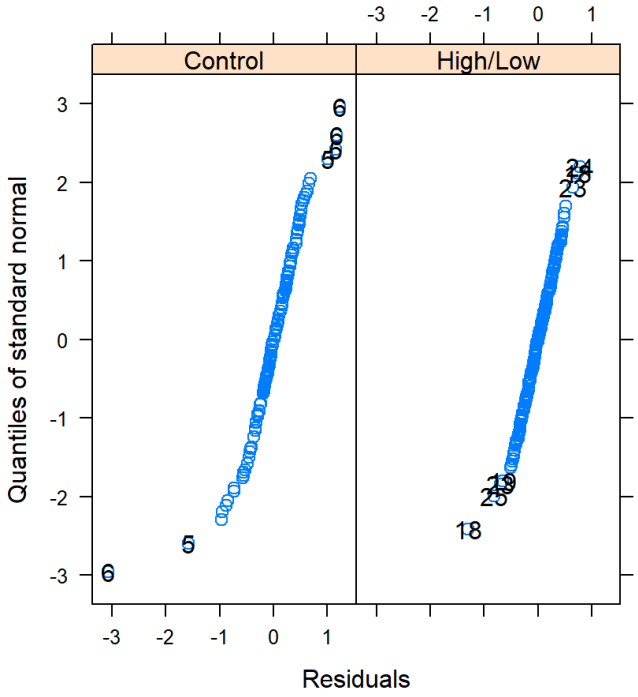
```
attach(ratpupcsv)
res <- resid(Model.FB)
ratred <- data.frame(ratpupcsv, res)
histogram(~res | factor(trtgrp), data = ratred, layout = c(2, 1), aspect = 2, xlab = "Residual",
    strip = strip.custom(factor.levels = c("Control", "High/Low")))
```




## 6.2 Quantile-quantile plots

Hide

```
attach(ratpupcsv)
qqnorm(Model.FB, ~resid(.) | factor(trtgrp), layout = c(2, 1), aspect = 2, id = 0.05,
    strip = strip.custom(factor.levels = c("Control", "High/Low")))
```

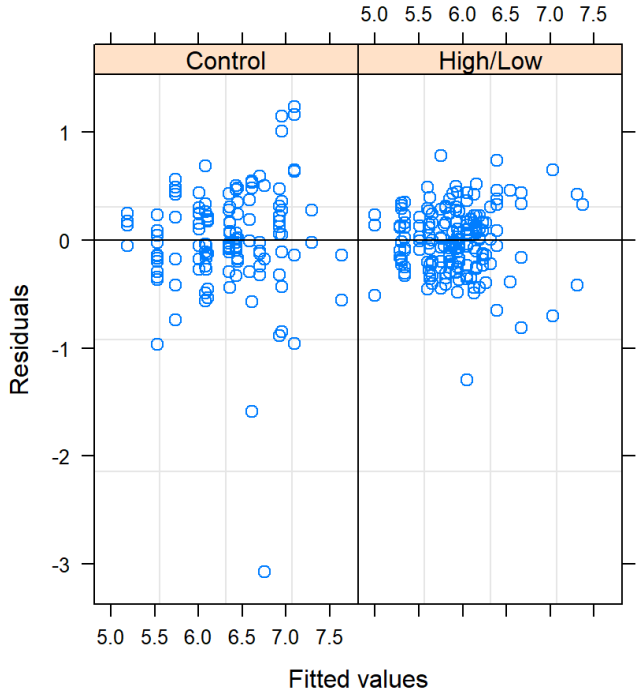



## 6.3 Homogeneity of variance plot

- fitted values vs. residuals
- the plot shows that residual variance is similar for small and large fitted values

Hide

```
attach(ratpupcsv)
plot(Model.FB, resid(.) ~ fitted(.) | factor(trtgrp), layout = c(2, 1), aspect = 2,
    abline = 0, strip = strip.custom(factor.levels = c("Control", "High/Low")))
```




## 6.4 Post-hoc test between all treatment levels.

Hide

```
library(multcomp)
Model.FB <- lme(weight ~ treatment + sex + litsize, random = ~1 | litter, method = "REML",
    ratpupcsv, weights = varIdent(form = ~1 | trtgrp))
summary(glht(Model.FB, linfct = mcp(treatment = "Tukey")))
```

```
     Simultaneous Tests for General Linear Hypotheses

Multiple Comparisons of Means: Tukey Contrasts

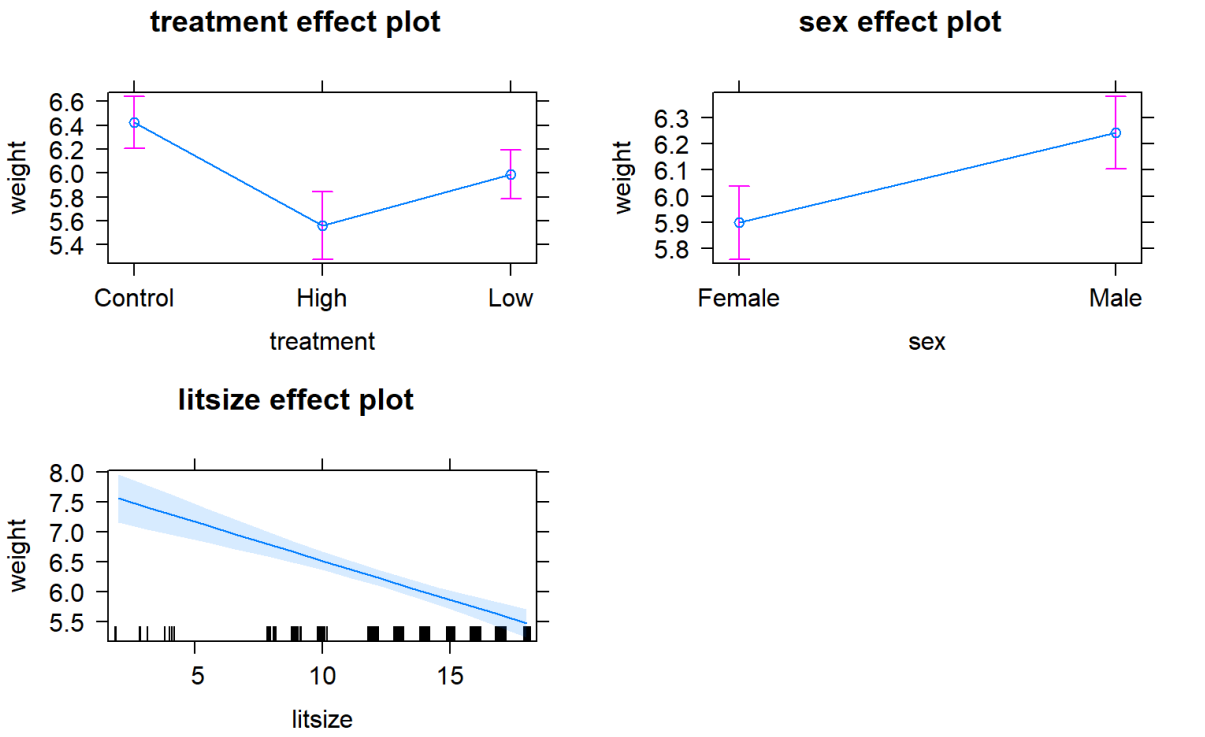

Fit: lme.formula(fixed = weight ~ treatment + sex + litsize, data = ratpupcsv,
    random = ~1 | litter, weights = varIdent(form = ~1 | trtgrp),
    method = "REML")

Linear Hypotheses:
                   Estimate Std. Error z value Pr(>|z|)
High - Control == 0  -0.8623     0.1829  -4.714   <0.001 ***
Low - Control == 0   -0.4337     0.1523  -2.848   0.0120 *
Low - High == 0       0.4286     0.1755   2.442   0.0384 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```

## 6.5 The effect plot

Hide

```
plot(allEffects(Model.FB))
```




# 7 Linear Mixed-Effect Model: Package lme4

Hide

```
lmer.model <- lmer(weight ~ treatment + sex + litsize + treatment * sex + (1 | litter),
    REML = FALSE, ratpupcsv)
summary(lmer.model)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: weight ~ treatment + sex + litsize + treatment * sex + (1 | litter)
   Data: ratpupcsv

     AIC      BIC   logLik deviance df.resid
   395.8    429.8   -188.9    377.8      313

Scaled residuals:
    Min      1Q  Median      3Q     Max
-7.5188 -0.5045  0.0254  0.5880  3.0233

Random effects:
 Groups   Name        Variance Std.Dev.
 litter   (Intercept) 0.0807   0.2841
 Residual             0.1617   0.4021
Number of obs: 322, groups:  litter, 27

Fixed effects:
                     Estimate Std. Error        df t value Pr(>|t|)
(Intercept)           7.91056    0.25784  41.07091  30.680  < 2e-16 ***
treatmentHigh        -0.79972    0.18207  40.29948  -4.392 7.92e-05 ***
treatmentLow         -0.38343    0.14911  35.83419  -2.572   0.0144 *
sexMale               0.41055    0.07273 299.07067   5.645 3.84e-08 ***
litsize              -0.12821    0.01755  38.34969  -7.305 9.05e-09 ***
treatmentHigh:sexMale  -0.11001    0.13083 308.82323  -0.841   0.4011
treatmentLow:sexMale   -0.08414    0.10502 302.81404  -0.801   0.4236
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmntHgh -0.580
treatmentLw -0.335  0.433
sexMale     -0.165  0.234  0.286
litsize     -0.911  0.372  0.045 -0.001
trtmntHgh:M  0.112 -0.382 -0.160 -0.556 -0.022
trtmntLw:sM  0.149 -0.176 -0.368 -0.693 -0.037  0.386
```

Hide

```
model1.fit.lmer <- lmer(weight ~ treatment + sex + litsize + treatment * sex + (1 |
    litter), ratpupcsv, REML = T)
summary(model1.fit.lmer)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: weight ~ treatment + sex + litsize + treatment * sex + (1 | litter)
   Data: ratpupcsv

REML criterion at convergence: 401.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-7.4725 -0.5001  0.0291  0.5735  3.0096

Random effects:
 Groups   Name        Variance Std.Dev.
 litter   (Intercept) 0.09652  0.3107
 Residual             0.16349  0.4043
Number of obs: 322, groups:  litter, 27

Fixed effects:
                     Estimate Std. Error        df t value Pr(>|t|)
(Intercept)           7.91165    0.27496  33.70992  28.773  < 2e-16 ***
treatmentHigh        -0.79903    0.19429  32.88474  -4.112 0.000245 ***
treatmentLow         -0.38317    0.15967  29.56842  -2.400 0.022919 *
sexMale               0.41169    0.07315 295.30143   5.628 4.24e-08 ***
litsize              -0.12838    0.01875  31.79751  -6.846 9.94e-08 ***
treatmentHigh:sexMale  -0.10702    0.13176 303.74417  -0.812 0.417291
treatmentLow:sexMale   -0.08387    0.10568 298.58691  -0.794 0.428077
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) trtmnH trtmnL sexMal litsiz trtH:M
treatmntHgh -0.581
treatmentLw -0.336  0.434
sexMale     -0.155  0.221  0.269
litsize     -0.910  0.372  0.045 -0.001
trtmntHgh:M  0.105 -0.360 -0.150 -0.555 -0.020
trtmntLw:sM  0.141 -0.166 -0.345 -0.692 -0.036  0.385
```

Hide

```
anova(model1.fit.lmer, type = "marginal")
```

```
Marginal Analysis of Variance Table with Satterthwaite's method
              Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
treatment     2.8416  1.4208     2  31.278  8.6905 0.0009965 ***
sex           5.1778  5.1778     1 295.301 31.6708 4.244e-08 ***
litsize       7.6619  7.6619     1  31.798 46.8652 9.937e-08 ***
treatment:sex 0.1522  0.0761     2 302.303  0.4656 0.6282199
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 7.1 Hypthesis Test 01 : with lem4

Hide

```
rand(model1.fit.lmer)
```

```
ANOVA-like table for random-effects: Single term deletions

Model:
weight ~ treatment + sex + litsize + (1 | litter) + treatment:sex
             npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>          9 -200.55 419.10
(1 | litter)    8 -245.25 506.51 89.406  1  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hide

```
Anova(model1.fit.lmer, type = 2, test.statistic = "Chisq")
```

```
Analysis of Deviance Table (Type II Wald chisquare tests)

Response: weight
               Chisq Df Pr(>Chisq)
treatment     23.3610  2  8.457e-06 ***
sex           56.9064  1  4.571e-14 ***
litsize       46.8652  1  7.604e-12 ***
treatment:sex  0.9312  2     0.6278
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hide

```
model2.fit.lmer <- lmer(weight ~ treatment + sex + litsize + (1 | litter), ratpupcsv,
    REML = F)
model3.fit.lmer <- lmer(weight ~ treatment + sex + (1 | litter), ratpupcsv, REML = F)
model4.fit.lmer <- lmer(weight ~ treatment + litsize + (1 | litter), ratpupcsv, REML = F)
model5.fit.lmer <- lmer(weight ~ sex + litsize + (1 | litter), ratpupcsv, REML = F)
anova(model1.fit.lmer, model2.fit.lmer)  #testing interaction effect
```

```
Data: ratpupcsv
Models:
model2.fit.lmer: weight ~ treatment + sex + litsize + (1 | litter)
model1.fit.lmer: weight ~ treatment + sex + litsize + treatment * sex + (1 | litter)
                Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
model2.fit.lmer  7 392.79 419.21 -189.39   378.79
model1.fit.lmer  9 395.81 429.78 -188.91   377.81 0.9721      2      0.615
```

Hide

```
anova(model2.fit.lmer, model3.fit.lmer)  #testing litsize effect
```

```
Data: ratpupcsv
Models:
model3.fit.lmer: weight ~ treatment + sex + (1 | litter)
model2.fit.lmer: weight ~ treatment + sex + litsize + (1 | litter)
                Df    AIC    BIC  logLik deviance Chisq Chi Df Pr(>Chisq)
model3.fit.lmer  6 422.98 445.62 -205.49   410.98
model2.fit.lmer  7 392.79 419.21 -189.39   378.79 32.19      1  1.398e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hide

```
anova(model2.fit.lmer, model4.fit.lmer)  #testing sex effect
```

```
Data: ratpupcsv
Models:
model4.fit.lmer: weight ~ treatment + litsize + (1 | litter)
model2.fit.lmer: weight ~ treatment + sex + litsize + (1 | litter)
                Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
model4.fit.lmer  6 442.05 464.69 -215.02   430.05
model2.fit.lmer  7 392.79 419.21 -189.39   378.79 51.262      1  8.084e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hide

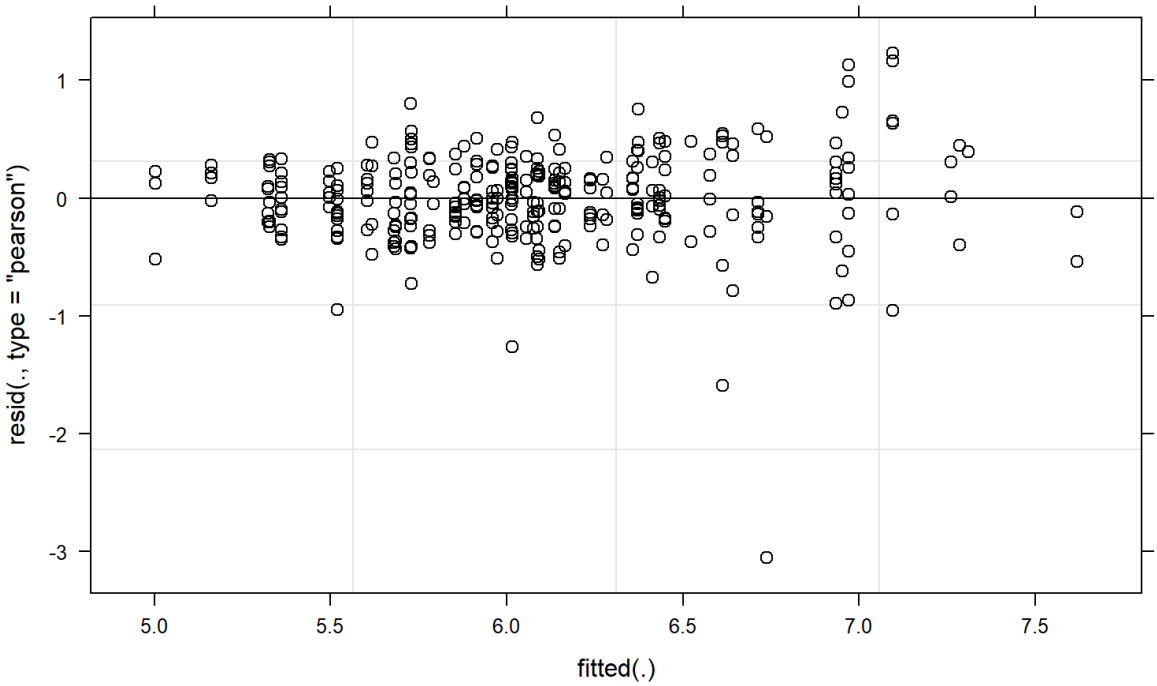
```
anova(model2.fit.lmer, model5.fit.lmer)  #testing treatment effect
```

```
Data: ratpupcsv
Models:
model5.fit.lmer: weight ~ sex + litsize + (1 | litter)
model2.fit.lmer: weight ~ treatment + sex + litsize + (1 | litter)
                Df    AIC    BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
model5.fit.lmer  5 407.48 426.35 -198.74   397.48
model2.fit.lmer  7 392.79 419.21 -189.39   378.79 18.695      2  8.716e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 7.2 check residuals for homoscedasticity

Hide

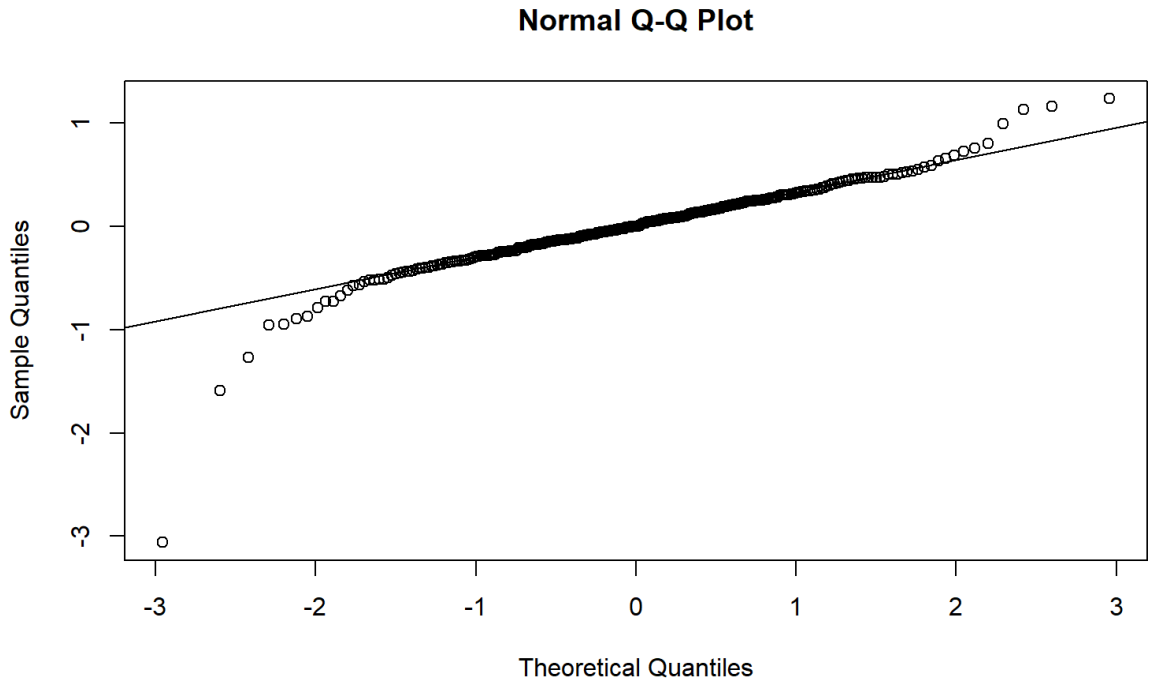
```
plot(model2.fit.lmer, col = "black")
```




## 7.3 check residuals for normality

Hide

```
qqnorm(resid(model2.fit.lmer))
qqline(resid(model2.fit.lmer))
```




Hide

```
model6.fit.lmer <- lmer(weight ~ treatment + sex + litsize + (1 | litter), ratpupcsv,
    REML = T)
summary(glht(model6.fit.lmer, linfct = mcp(treatment = "Tukey")))
```

```
     Simultaneous Tests for General Linear Hypotheses

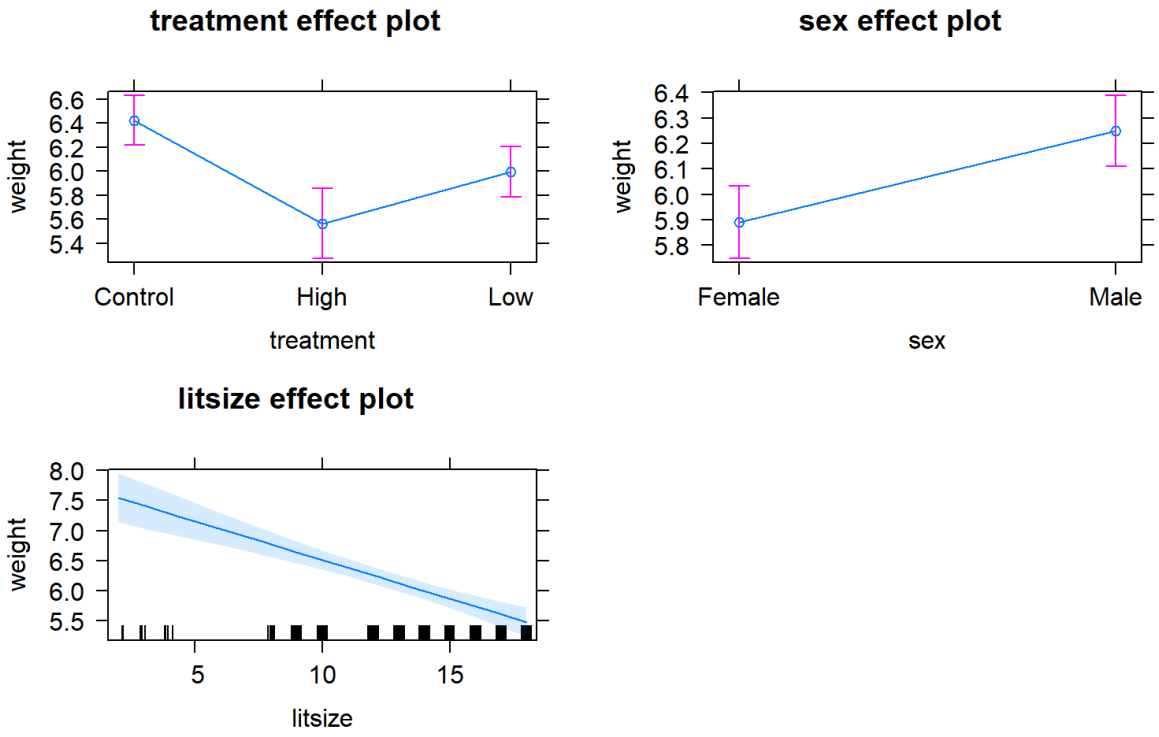
 Multiple Comparisons of Means: Tukey Contrasts


 Fit: lmer(formula = weight ~ treatment + sex + litsize + (1 | litter),
     data = ratpupcsv, REML = T)

 Linear Hypotheses:
                     Estimate Std. Error z value Pr(>|z|)
 High - Control == 0  -0.8587     0.1818  -4.723   <0.001 ***
 Low - Control == 0   -0.4285     0.1504  -2.849   0.0119 *
 Low - High == 0       0.4302     0.1804   2.385   0.0446 *
 ---
 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
 (Adjusted p values reported -- single-step method)
```

Hide

```
plot(allEffects(model6.fit.lmer))
```




# Reference

Crawley, Michael J. 2013. “The R Book Second Edition.” John Wiley & Sons.

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